Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the two positivity conjectures for Morin singularities. The infrastructure computes $\mathcal{Q}_d$ for $d \leq 7$ from the explicit orbit equations of Bérczi–Szenes §7.3, self-checked against every invariant fixed in advance and agreeing with Rimányi's published tables wherever those reach. On top of it: Rimányi's conjecture is verified for A_6 (first time), the two reductions of the A_5 handoff note are shown to survive at $d = 6$ but to *fail* at $d = 7$ – while Rimányi's conjecture itself survives there, so the reduction machinery, not the conjecture, is what is specific to small d — the proposed prefix-positivity route is shown to fail already at $d = 6$, and a notion of *denominator certificate* is introduced whose successes are proofs and whose failures are theorems. The certificate machinery reproduces the $d = 4$ strong positivity theorem and Proposition 3 of the handoff note mechanically. All coefficients agree with Rimányi's independently computed tables wherever those reach, including A_7. Notation follows `rimanyi_positivity.pdf`, `report.pdf` and `a5_weak_positivity_handoff.pdf`, cited as [N], [R], [H].

1 Setting and notation

For the Morin singularity A_d, the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d-dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\widehat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2\, \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) \;=\; \frac{\prod_{m<l}(1 - z_m/z_l)\, \mathcal{Q}_d}{\prod_{m+r\leq l}(1 - z_m/z_l - z_r/z_l)} \;=\; \frac{N_d(x)}{\prod_r (1 - f_r(x))} \;=\; \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ-free Chern coefficient of [R, Thm. 5].

2 Infrastructure

2.1 Computing $\mathcal{Q}_d$

We take the route of [N, §7.3]: rather than parametrise the orbit and eliminate the group variables, we write down the quadratic relations of Proposition 7.3 directly, saturate by the defect-zero coordinates to remove the spurious components that appear from $d = 6$ on ([N, §7.4]), and read the multidegree off an initial ideal.

The design point worth stating is that every quantity the computation can be checked against is known in advance, and all of them are checked: multihomogeneity of each relation under the torus weights; the codimension $\dim \widehat{N}_d - \binom{d}{2}$ forced by homogeneity of the residue formula; vanishing on random B_d-translates of ϵ_{ref}; $\deg \mathcal{Q}_d$; exactness of the chamber division; and the normalisation fixing the c_1^d coefficient. The stage refuses to emit an artifact otherwise. Externally, the resulting Thom polynomials agree with Rimányi's published tables at every order and relative dimension those cover (§4).

2.2 Reaching $d = 7$

$d = 7$ is 34 variables against 22, and on the obvious configuration it does not finish at all. The cost is concentrated in one place — the Gröbner basis, which the saturation invokes once per defect-zero coordinate — and two choices govern it.

First, `std` rather than `slimgb`, and the defect-zero variables placed last in degrevlex. On the 21 quadrics of A_7 over GF(32003) the four combinations differ by more than two orders of magnitude:

Table 1: Groebner basis of the unsaturated A_7 ideal. Same ideal, four configurations.

variable order	algorithm	time	basis size
as generated	`slimgb`	31.2 s	467
as generated	`std`	1.4 s	467
defect-zero last	`slimgb`	2.2 s	224
defect-zero last	`std`	**0.2 s**	**224**

Second, and less obviously, the *sequence* of saturations. Saturating by a variable is cheap if that variable is last in degrevlex — a basis then saturates on dividing each element by the largest power of the variable dividing it, replacing an elimination with one basis computation. But moving a coordinate to the very end perturbs the order that made the basis cheap. Saturating the defect-zero coordinates *back-to-front* keeps each move small; front-to-back does not, and at $d = 7$ its first step alone fails to finish. Back-to-front, all twelve complete in seconds.

Third, the variable order *within* each block. Sorting by the level l descending, so that the deepest coordinates sit where degrevlex does least work, takes the twelve saturating bases from 14.0 s to 2.7 s. Sorting by the defect $l - m - r$ instead gives 4.2 s and ascending l gives 15.6 s, so it is the direction that matters rather than merely having a rule. Two orders we could not time at all turned out to be an artefact of moving generators between rings by printing and reparsing them, which at $d = 7$ exceeds Python's recursion limit; permuting exponent vectors instead is exact and removed a further third of the cost.

Fourth, the two phases after the basis are combinatorics, not algebra. The initial ideal is monomial, and the minimal primes of a monomial ideal are generated by variables: P_S contains the ideal exactly when S meets the support of every generator. They are therefore the minimal *transversals* of those supports, which a bounded incremental search over integer bitmasks finds in about two seconds where a general-purpose primary decomposition takes 41 s. Bounding the search at the expected codimension is sound — every subset of a small transversal is itself small — and leaves the codimension check intact in both directions. The multiplicity of a component is then the number of monomials outside an artinian monomial ideal, and the boxes are small enough — the largest at $d = 7$ holds 24 monomials — that enumerating one outright beats asking Singular for the length by a factor of 150, on every one of the 51 distinct localisations.

Table 2: Where the time goes at $d = 7$, over $\mathbb{Q}$. Every artifact is byte-identical before and after.

phase	before	after	what changed
saturation (12 bases)	63.4 s	3.6 s	`std`, order by level, exponent remapping
final basis	0.6 s	0.6 s	—
minimal primes	40.6 s	2.0 s	transversals over bitmasks
multiplicities	33.1 s	0.9 s	staircase enumeration, not Singular
weight products	—	1.6 s	—
total	**138 s**	**9.1 s**	

Two things did *not* help, and are worth recording so they are not tried again. The base field is irrelevant: $\mathbb{Q}$, GF(32003) and GF($2^{31}-1$) all take within 4% of each other and produce byte-identical initial ideals, so coefficient growth is not what this computation is spending on. And none of the twelve saturations can be skipped: after each one we tested whether the ideal was already saturated with respect to any remaining defect-zero coordinate — a condition checkable on the initial ideal alone, since $\mathrm{in}(I) : v = \mathrm{in}(I)$ forces $I : v = I$ — and the answer was never once yes.

Reordering the variables changes the degeneration, not the answer: $d = 7$ goes from 621 components to 572, with different multiplicities, summing to the same $\mathcal{Q}_7$. The component structure is an artefact of the term order; the multidegree is not.

2.3 Scale of the objects

Table 3 is the practical reason sampling is no substitute for a proof: roughly half the coefficients of N_d are negative at every order, and the counts grow fast.

Table 3: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \widehat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure.

	vars	$\dim \widehat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	N_d terms	$\deg N_d$	N_d negative
A_4	3	7	1	3	54	13	27 (50%)
A_5	4	13	3	19	814	31	414 (51%)
A_6	5	22	7	418	26618	66	13309 (50%)
A_7	6	34	13	14586	1210404	122	605234 (50%)

2.4 Exactness

No result below depends on floating point. One point deserves emphasis because it is a trap: Laurent coefficients are accumulated in `int64`, and they outgrow it. A wrapped accumulator reads as a large *negative* number, that is, as a spurious counterexample to the very conjecture under test. The evaluator therefore replays the fixed point in floating point as a magnitude probe and raises rather than returning wrapped values. Beyond that point the expansion is run in residue arithmetic instead; see §3.

3 Results for A_6

Proposition 1 (verified)**.** *Rimányi's conjecture holds for A_6 at every relative dimension $\ell \leq 7$, and for A_5 at $\ell \leq 11$.*

These are finite exact computations, not proofs of the infinite statement. For A_5 this matches the range of [R, Thm. 8]; the A_6 verification is new. Past $\ell = 5$ the coefficients outgrow `int64` — they reach 1.75×10^{22} at A_6, $\ell = 7$ — so the expansion is run modulo several word-sized primes and reconstructed by CRT, with one prime held back and spent on verifying the reconstruction. The grouping into Chern monomials is computed from the grid geometry rather than from which cells are nonzero, so that the residues of a given coefficient correspond across primes; grouping by value would misalign them, since a nonzero coefficient can vanish modulo one prime and not another.

3.1 External agreement

Rimányi's tables at `tpp.web.unc.edu` give the Thom polynomials at relative dimension one, computed by the restriction-equation method. All 15, 30 and 58 coefficients for A_4, A_5, A_6

agree with ours. Since that method is independent of the residue formula implemented here, this constrains $\mathcal{Q}_d$ externally rather than self-consistently, and is a considerably sharper check than the eleven classical values at $\ell = 0$.

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 4. The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §3.3.

Table 4: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	$(1, 1, 2, 1)$	-1	0
A_6	796	10	$(1, 1, 2, 1, 0)$	-2	2

3.2 The reductions of [H] at $d = 6$

Observation 2. *Both reductions of [H] survive at $d = 6$ throughout the tested range (*$\deg \le 11$*): the unpaired tail $i > j \Rightarrow A_{i,j,\ldots} \ge 0$, and the paired inequality $A_{i,j,\ldots} + A_{j-i,j,\ldots} \ge 0$ for $0 \le i \le j$.*

This is evidence that the τ-pairing frame is structural rather than an A_5 accident.

3.3 Prefix positivity fails at $d = 6$

[H, §8] observes that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and [H, §11.4] proposes proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \le i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. *$F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.*

Proof. The prefix sum runs over $r \le i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient with $i = 0$ survives untouched. F_5 has no negative coefficient with $i = 0$ in the tested range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$. Confirmed along two independent code paths, truncated series products in exact integers and the XLA fixed-point grid. □

The route of [H, §11.4] is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

4 A_7: the reductions break, the conjecture does not

$\mathcal{Q}_7$ is computable — see §2.2 — and it changes the picture. It is worth being precise about what fails, because the two are easy to conflate. Strong Laurent positivity has been false since $d = 5$ and is no more false at $d = 7$. Rimányi's conjecture, the actual target, *survives*: every Chern coefficient of Tp_{A_7} is positive at every relative dimension we reach. What breaks is the *reduction machinery* of [H] — the two inequalities through which the A_5 programme runs.

4.1 Validation first

Because the claim is a negative one, the input has to be beyond question. Our Tp_{A_7} reproduces Rimányi's published tables exactly: all 15 coefficients at $\ell = 0$ and all 105 at $\ell = 1$, and likewise for A_4, A_5, A_6. Those tables are maintained by Rimányi himself at `tpp.web.unc.edu` and are computed by the restriction-equation method — an approach independent of the residue formula implemented here — so the agreement constrains $\mathcal{Q}_7$ externally rather than self-consistently.

We note the limit of that check. The companion page `tpp.web.unc.edu/tp-linfinity/` tabulates the rational functions F_Q, whose numerators are the closest published analogue of our $\mathcal{Q}_d$; for A_4 it gives numerator $2z_1 + z_2 - z_4$, exactly our $\mathcal{Q}_4$. But that page stops at $\mu = 6$: **no** A_7 **data appears there**, so $\mathcal{Q}_7$ itself has no published counterpart to compare against. The Thom polynomial tables are therefore the strongest external constraint available, and they are what we use.

Internally, every offending coefficient below agrees between two unrelated code paths — the truncated series in exact Python integers and the XLA fixed-point grid — and all of them sit at total degree 7 to 10, inside the truncation degree, so they are exact rather than artefacts of a cut-off.

4.2 How the reductions fail

Observation 4. *The unpaired tail fails at $d = 7$: $A_{(2,1,1,2,2,1)} = -1$ with $i = 2 > j = 1$. Proposition 3 of [H] proves this cannot happen at $d = 5$, and it does not happen at $d = 6$.*

Observation 5. *The paired inequality fails at $d = 7$, and does so in the sharpest possible way. Take $\beta = (1, 2, 1, 2, 2, 1)$. Then $j = 2i$, so $\tau(\beta) = \beta$: this is a* fixed point *of the transposition, where the paired inequality reads $2A_\beta \geq 0$. But $A_\beta = -1$. There is no partner to appeal to.*

4.3 Why Rimányi's conjecture survives it

The mechanism is the one that saved $d = 5$, and it is worth seeing at $d = 7$ concretely. In the α coordinates $\beta = (1, 2, 1, 2, 2, 1)$ is $\alpha = (1, 1, -1, 1, 0, -1, -1)$, so it contributes to the Chern monomial indexed by the multiset $M = \{1, 1, 1, 0, -1, -1, -1\}$ — at $\ell = 0$, the monomial $c_1c_2^3$. That multiset has 35 ballot orderings, of which eight carry a nonzero coefficient:

Table 5: The Chern coefficient $C(M)$ for $M = \{1, 1, 1, 0, -1, -1, -1\}$ at $d = 7$. The offending β of the second observation is the third row.

α	β	A_β
$(1, -1, 1, 1, 0, -1, -1)$	$(1, 0, 1, 2, 2, 1)$	1
$(1, 0, 1, 1, -1, -1, -1)$	$(1, 1, 2, 3, 2, 1)$	4
$(1, 1, -1, 1, 0, -1, -1)$	$(1, 2, 1, 2, 2, 1)$	-1
$(1, 1, 0, 1, -1, -1, -1)$	$(1, 2, 2, 3, 2, 1)$	-4
$(1, 1, 1, -1, -1, -1, 0)$	$(1, 2, 3, 2, 1, 0)$	5
$(1, 1, 1, -1, -1, 0, -1)$	$(1, 2, 3, 2, 1, 1)$	6
$(1, 1, 1, -1, 0, -1, -1)$	$(1, 2, 3, 2, 2, 1)$	6
$(1, 1, 1, 0, -1, -1, -1)$	$(1, 2, 3, 3, 2, 1)$	18
	$C(M)$	**35**

Five negative units against forty positive: $C(M) = 35 > 0$. And 35 is precisely the coefficient of $c_1c_2^3$ in Rimányi's published Tp_{A_7} at $\ell = 0$, so the cancellation is confirmed against his table and not only against our own arithmetic. The unpaired-tail violation behaves the same way: $A_{(2,1,1,2,2,1)} = -1$ sits inside $M = \{2, 1, 0, 0, -1, -1, -1\}$, the monomial $c_1^2c_2c_3$, whose coefficient is 301.

So $d = 7$ separates two things that $d \leq 6$ kept together. The conjecture is about $C(M)$, and $C(M)$ is fine. The reductions are about the individual A_β and their τ-orbits, and those are not. The τ-pairing is a way of certifying the sum by controlling its summands two at a time; at $d = 7$ the summands are no longer controllable that way, while the sum is unharmed.

4.4 Consequences

The τ-pairing frame is therefore $d \le 6$, not structural. Anything built on it — the unpaired-tail reduction, the paired series $\mathcal{P}$ of [H, eq. (12)], the $J_d(1/2,\cdot)$ analysis of §6 — is an argument about small d and cannot be the shape of a general proof. That is a strong constraint on where to look next, and it is the main reason we report the failure rather than the verification.

5 Denominator certificates

Definition 6. Let $N/\prod_{r=1}^{R}(1-f_r)$ with each f_r nonnegative and of zero constant term. An *order-k certificate* is an identity

$$N \;=\; \sum_{|S|\le k} P_S \prod_{r\in S}(1-f_r), \qquad \text{every } P_S \text{ coefficientwise nonnegative,}$$

over subsets $S \subseteq \{1,\dots,R\}$.

Dividing through, $F = \sum_S P_S/\prod_{r\notin S}(1-f_r)$ is a sum of products of nonnegative series, so $F \ge 0$. Order 0 is the statement that N itself is nonnegative. [H, §10.3] reports a first-order search of this shape coming back infeasible.

5.1 Why a returned certificate is a proof

$P_\emptyset$ is not a free unknown: since $\prod_{r\in\emptyset}(1-f_r)=1$, it is the remainder $N-\sum_{S\ne\emptyset} P_S\prod_{r\in S}(1-f_r)$. We therefore solve a feasibility LP in the nonempty parts only, snap the solver's floating-point answer to exact rationals, and recompute $P_\emptyset$ in exact arithmetic. The identity then holds by construction, and the single remaining question, whether that exact remainder is coefficientwise nonnegative, is decided exactly. The LP is demoted to a search heuristic; correctness never depends on it. The output is a finite algebraic identity checkable in any computer algebra system.

5.2 Why a failure is a theorem

Lemma 7 (projection)**.** *A constraint on a monomial of total degree T involves only the coefficients of the P_S of degree $\le T$, because every $\prod_{r\in S}(1-f_r)$ has constant term 1 and no negative-degree terms. Truncating both the constraints and the unknowns at degree T is therefore an exact* projection *of the feasible set, not a further restriction.*

Consequently infeasibility of the truncated program proves that no certificate of that order exists at *any* degree. Failed searches become impossibility results.

5.3 A_4

Proposition 8. *F_4 admits an order-4 certificate with* $\deg P_S \le 8$*, exhibited explicitly and verified exactly; and no certificate of order ≤ 3 exists at any degree. The minimal order is* 4.

This is an independent computational proof of [N, Thm. 1] by a method using no special feature of $d=4$. The lower bound already follows by hand at depth 1: writing m_S for the constant term of P_S, matching the constant term of $N = 1-a-b-c+\cdots$ forces $\sum_S m_S \le 1$, while matching the coefficients of a, b, c forces

$$\sum_{S\ni f_0} m_S \ge \tfrac{1}{2}, \qquad \sum_{S\ni f_2} m_S \ge 1, \qquad \sum_{S\ni f_5} m_S \ge 1,$$

where $f_0 = 2a$, $f_2 = b+ab$, $f_5 = c+abc$ are the only factors containing the respective bare monomials. With a budget of 1 these can only be met if a single S contains all three. The machine-found certificate saturates all four bounds exactly, its only two constant terms being $\frac{1}{2}(1-f_0)(1-f_2)(1-f_4)(1-f_5)$ and $\frac{1}{2}(1-f_2)(1-f_4)(1-f_5)(1-f_6)$.

5.4 A_5: how far the obstruction reaches

Proposition 9. *No denominator certificate of order ≤ 7 exists at any degree for F_5, nor for $F_5/(1-a)$.*

For F_5 this is expected, since $F_5 \not\geq 0$, and serves as a control that the engine never manufactures a certificate for a false statement. For the prefix series, which *is* coefficientwise nonnegative as far as we compute, it is informative: it extends [H, §10.3] from order 1 to order 7. The bound sharpens with probe depth: depth 1 gives ≥ 4, depth 2 gives ≥ 7, depth 3 rules out everything up to 7.

6 The unpaired tail

6.1 One series

Set $H(a) = (1-a)/(1-2a)$ and $J_d = F_d/H$. After H is removed, every occurrence of a in a numerator or denominator monomial is accompanied by b: the chamber monomials containing a are $z_1/z_l = ab\cdots$, and only $z_1/z_2 = a$ is bare, which is exactly the factor $1-2a$ that H removes. Hence $\deg_a[b^j c^k \cdots] J_d \le j$, and convolving with $H(a) = 1 + \sum_{n\ge1} 2^{n-1}a^n$ gives, for $i > j$,

$$A_{i,j,k,\dots} \;=\; 2^{i-1}\,[b^j c^k \cdots]\,J_d(1/2, b, c, \dots).$$

The entire unpaired tail, infinitely many coefficients, is therefore one positivity question about a single series in $d-2$ variables. This is [H, Lemma 2 and Prop. 3] for $d=5$, stated so that it applies verbatim for any d.

Our $J_5(1/2, b, c, e)$ agrees with [H, eq. (9)] coefficient for coefficient to degree 12, which is what validates the construction. $J_6(1/2, b, c, d, e)$ is coefficientwise nonnegative as far as we compute (deg ≤ 10).

6.2 Multiplicative certificates

The two successful proofs in the literature, [N, Thm. 1] and [H, Prop. 3], are *multiplicative*: products of ratios $(1-u)/(1-v)$, each nonnegative by Lemma 1 because $v - u \geq 0$. That shape is not a finite sum $\sum_S P_S \prod_{r\in S}(1-f_r)$ with $P_S \geq 0$, so the additive certificates of §4 structurally cannot find them.

The additive search is obstructed on $J_5(1/2, \cdot)$, whose nonnegativity is Proposition 3, so the additive class is not the one to use here.

Presenting the numerator in factored form, the Vandermonde part being already a factorisation, one can search for a matching of numerator factors to denominators satisfying the Lemma 1 condition, cancel whatever divides exactly, and ask whether what remains is nonnegative.

Proposition 10. *For $d = 5$ this closes: all nine numerator factors pair off, three denominators cancel exactly, and the leftover numerator is exactly* 1*. Proposition 3 is reconstructed mechanically.*

For $d = 6$ the same search does not close, and §4 says why that is not worth pursuing: the unpaired tail is a statement about individual A_β, and at $d = 7$ it is false. The mechanisation is therefore a way of reproducing and checking the small-d arguments of [H], not a route to a general one.

7 Assessment

What the computations suggest about proof strategy:

1. The τ-pairing reduction is *not* the right frame in general. Both its inequalities survive at $d = 6$ and both fail at $d = 7$ (§4), while Rimányi's conjecture survives, so the machinery is specific to small d even though the statement is not. The prefix-positivity stepping stone fails one order earlier still.

2. Additive, Positivstellensatz-shaped certificates are obstructed on *every* target tried, including provable ones. On present evidence they are not the tool for this problem, and we would not recommend further effort extending [H, §10.3] in that direction.

3. Multiplicative certificates are the ones that succeed where anything does: they reproduce both published proofs from scratch. But what they certify — the individual A_β — is exactly what stops being controllable at $d = 7$, so they mechanise the small-d arguments rather than generalise them.

4. Since the τ-orbit is not the right unit of cancellation, the open question is what is. Table 5 is suggestive: the negative mass is small, bounded and spread over few orderings, while the positive mass is large and concentrated in one. A bound on the negative part of $C(M)$ in terms of its largest term would settle the conjecture without needing any pairing at all, and nothing we have computed argues against one existing.

8 Open questions and sharp edges

Points where these computations run past what we can justify from the literature. Each states what was checked, what it settles, and what it does not. Intended for discussion rather than as claims.

8.1 What $\mathcal{Q}_d$ is an invariant of

Take the corank-two 2-jet space $\mathrm{Hom}(\mathbb{C}^2, \mathrm{Sym}^2\,\mathbb{C}^2)$, of dimension 6, and the class $I_{2,2}$ with local algebra $\mathbb{C}[[x,y]]/(xy,\, x^2+y^2) \cong \mathbb{C}[[x,y]]/(x^2,y^2)$. Four germs of this one class give three different $B_2 \times B_2$-orbit closures (Table 6); the two standard presentations of the *same* algebra already disagree.

Table 6: Borel orbit closures of four germs of the single class $I_{2,2}$.

germ	orbit closure	codim
$(xy,\ x^2+y^2)$	q^u_{02}	1
$(x^2+xy,\ y^2)$	q^u_{02}	1
$(x^2,\ y^2)$	$q^u_{02},\ q^u_{11}$	2
$(y^2,\ x^2)$	three quadrics, determinantal	2

The natural repair — demand the representative be generic against the flags — fails, for a structural reason. On this space $GL_2 \times GL_2$ *does* have a dense orbit, so $I_{2,2}$ is the generic corank-two 2-jet and has codimension 0; but $B_2 \times B_2$ does not, its generic orbit being 5-dimensional in a 6-dimensional space. The action has modality at least one, the generic jet lies on a one-parameter family of orbits, and their closures are hypersurfaces whose equations move with the representative. There is no generic Borel orbit to single out.

This is not a defect in [BS]. Their $\mathcal{Q}_d$ is not "the class of the A_d locus"; it is the equivariant class of *one specific* Borel orbit closure, that of a canonically defined reference jet, and the residue formula is what converts it into a Thom polynomial. The choice of ϵ_{ref} is part of the theorem, not a normalisation one is free to vary.

Question. Is there a corank-two analogue of ϵ_{ref}, together with a residue formula proved against it? Does the fibration of [BS, Lemma 7.1] have a corank-two counterpart? Without both halves a corank-two multidegree is a number with no theorem attached, which is why `multidegree/corank2.py` stops there and emits no artifact.

8.2 The acting group beyond 2-jets

On 2-jets only the *linear* parts of a source and target diffeomorphism act: for f with no constant or linear term, $f(\varphi_1 + \varphi_2 + \cdots) = f(\varphi_1) + O(3)$ and $\psi(f) = \psi_1(f) + O(4)$. From order 3 the higher parts contribute, so the acting group is the jet group. Every $I_{a,b}$ with $b \geq 3$ lives in the b-jet space — $I_{2,3}$ and $I_{3,3}$ in dimension 14, $I_{2,4}$ in dimension 24 — so the whole family past $I_{2,2}$ needs this.

Question. What is the right Borel of the jet group, and does the defect-zero saturation trick of §2.2 have an analogue that isolates the orbit component?

8.3 Which positivity, in which basis

Rimányi's conjecture is a statement about the relative *Chern monomial* basis, and it is a corank-one statement. From Rimányi's registry, at relative dimension 0, $\mathrm{Tp}(I_{2,2}) = c_2^2 - c_1 c_3 = s_{2,2}$, the Giambelli–Thom–Porteous class of Σ^2. We expanded the published $I_{2,2}$, $I_{2,3}$, $I_{2,4}$, $I_{3,3}$ in the Schur basis: every one is Schur-positive, and every one has mixed signs in Chern monomials. So the I family tests Schur positivity — Pragacz–Weber — and refutes any naive extension of the Chern-monomial statement.

Question. Is Chern-monomial positivity special to Morin singularities, or the shadow of a statement about a basis interpolating between Chern monomials and Schur polynomials? The A_d Thom polynomials are positive in both; the $I_{a,b}$ only in one.

8.4 Why $\mathcal{Q}_8$ is not a matter of waiting

Between $d = 7$ and $d = 8$ the ambient dimension goes $34 \to 50$, deg $\mathcal{Q}_d$ goes $13 \to 22$, the basic equations $21 \to 39$ and the defect-zero coordinates $12 \to 16$. The whole of $d = 7$ now costs 9 s, and its first saturating basis 0.2 s. At $d = 8$ that same first basis did not finish in two hours, at 4 GB.

It is not a constant-factor problem, and the shape of it can be measured. The ideal is homogeneous, so a degree-bounded basis is exact up to its bound, and walking the bound upwards prices each degree (Table 7).

Table 7: Degree-bounded Gröbner basis of the 39 quadrics of A_8 over GF(32003).

degree bound	elements	time	ratio
5	752	0.9 s	—
6	1901	7.1 s	8.3×
7	4734	61.3 s	8.7×
8	10521	424.5 s	6.9×
9	19243	1737.6 s	4.1×

The bound that would be needed is the other half. The saturated basis reaches degree 3 at $d = 5$, 6 at $d = 6$ and 14 at $d = 7$, so $d = 8$ should need somewhere around 25. At a per-degree factor of 5, degree 25 is roughly 10^{11} times the cost of degree 9 — and memory, already 4 GB at degree 9, gives out long before. No amount of hardware closes that, and none of the levers we tried moves it: ten variable orders and two algorithms all fail to finish the first basis in 15 minutes; the base field is irrelevant; seeding the computation with the already-saturated $\mathcal{Q}_7$ ideal, which is legitimate since truncating jets maps $\mathcal{O}_8$ into $\mathcal{O}_7$, does not help either.

So the question is not how to speed this up. It is whether $\mathcal{Q}_8$ is needed. For Rimányi's conjecture at $d = 8$ it is not: his registry publishes Tp_{A_8} and Tp_{A_9} at relative dimension 0, computed by restriction equations rather than by a residue formula, and all 22 and 30 coefficients are positive. The conjecture is already supported two orders beyond where we can compute. What $\mathcal{Q}_8$ would buy is what the Chern coefficients cannot give: the individual A_β, and so the ability to ask at $d = 8$ the questions §4 answers at $d = 7$. That is a reason to want a different route to $\mathcal{Q}_d$ — localisation on a resolution, or the fibration of [N, Lemma 7.1] — not a faster Gröbner basis.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.